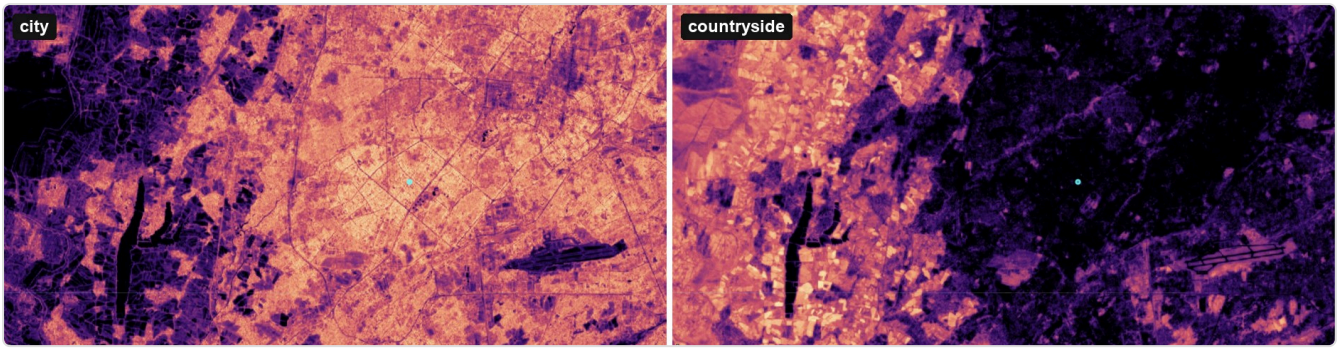
Day 09 tutorial: City or countryside? Ask the satellite

Theme: Urban–rural. Google's **AlphaEarth** model has looked at years of satellite images of every place on Earth and summed up each 10 × 10 m square in 64 numbers: its **embedding**. Places that look and behave alike get similar numbers. In this tutorial you pick one place in Tashkent and one place in the fields, and ask: *which other places are like this one?* You work in the **Google Earth Engine Code Editor**, in your browser. It takes about 30 minutes.

Printable PDF: English · Русский · O‘zbekcha



What you'll learn

- What a satellite **embedding** is, and why it's useful.
- How to run a script in Google Earth Engine, the free cloud tool used by scientists everywhere.
- How to do a **similarity search**: find every place that looks like an example you choose.

Part 0. Three ideas before you start

1. **A fingerprint for every place.** For each 10 m pixel and each year since 2017, AlphaEarth combined everything the satellites saw (Sentinel-2 colours, Sentinel-1 radar, Landsat and more) into 64 numbers. You can't read them one by one, but together they describe the place: roofs or crops, trees or water, how it changes over the seasons.
2. **Similar places have similar numbers.** To compare two places, multiply their 64 numbers pair by pair and add them up (a **dot product**). The answer is close to **1** for very similar places and lower for different ones.
3. **Earth Engine does the work in the cloud.** Tashkent alone has millions of pixels. Earth Engine computes them on Google's computers and only sends you the picture.

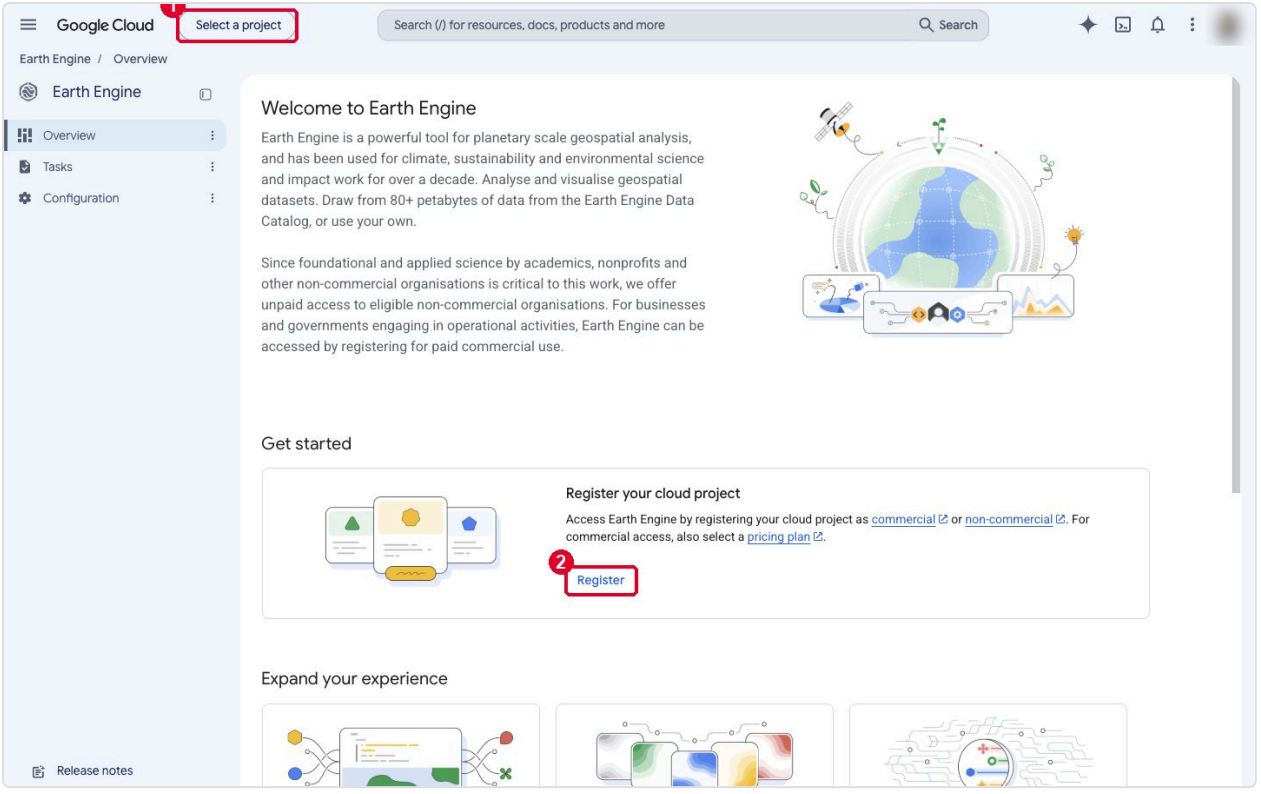
Part 1. Get access to Earth Engine (once, about 10 minutes)

Earth Engine is free for students, teachers and researchers, but you need a Google account and a **Cloud project** registered for **noncommercial** use. You do this only once.

1a. Open the Earth Engine page in Google Cloud

1. Sign in to your Google account, then open **console.cloud.google.com/earth-engine**.

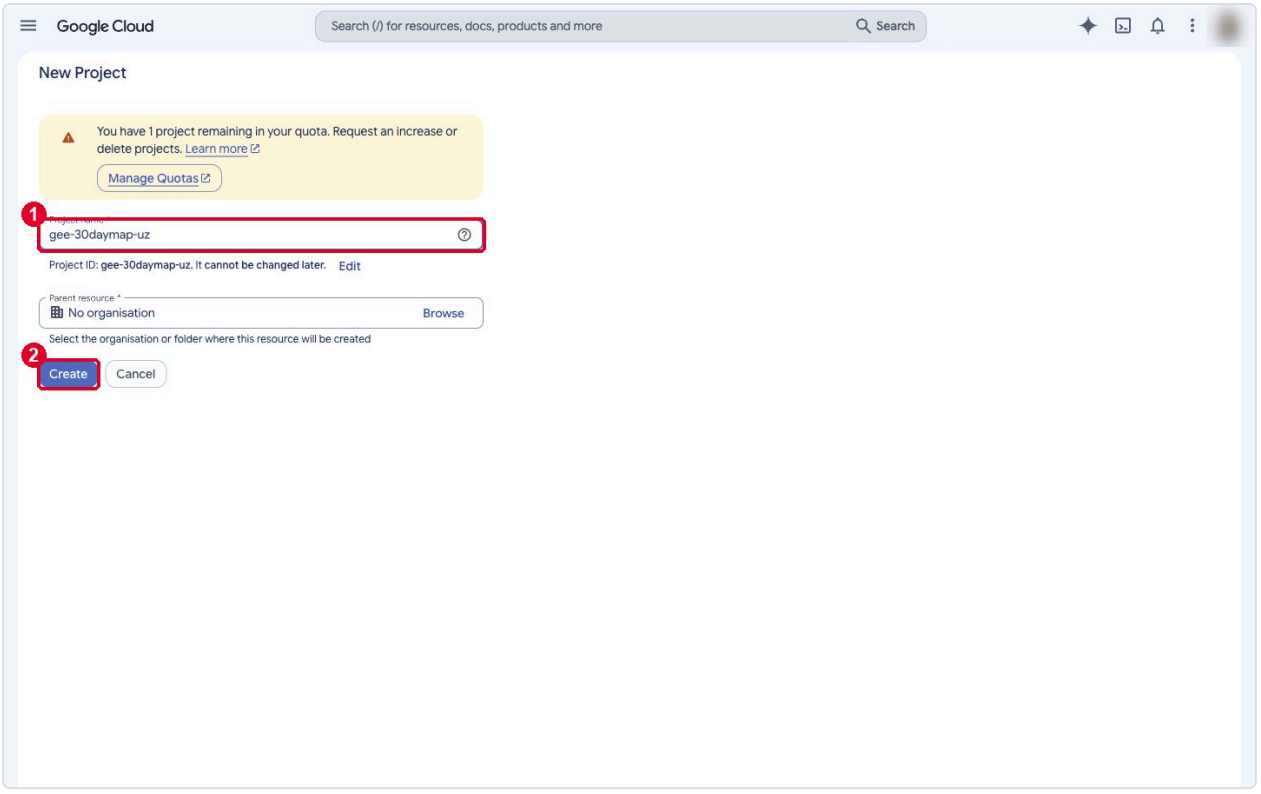
2. You see **Welcome to Earth Engine**. If you don't have a project yet, click **Select a project** at the top (**1**) → **New project**. If you already have one, choose it there and go straight to **Register** (**2**).



1b. Create a project

1. **Project name:** type a short name, for example `gee-30daymap-uz` (**1**). Google makes a **Project ID** from it, which you can't change later.

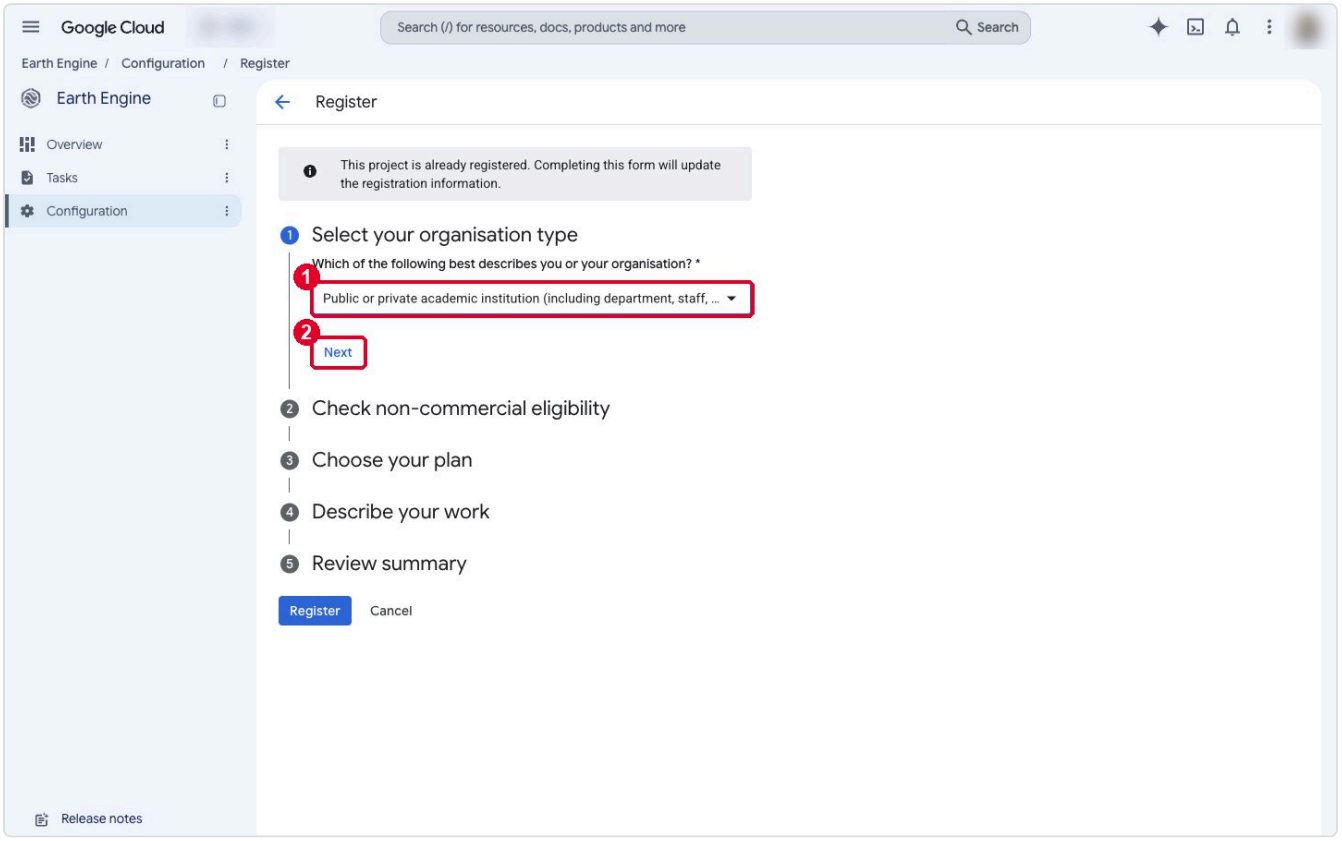
2. Leave **Parent resource** as **No organisation**, then click **Create** (**2**). Wait a few seconds until the project is ready, and make sure it's selected at the top of the page.



1c. Register the project for noncommercial use

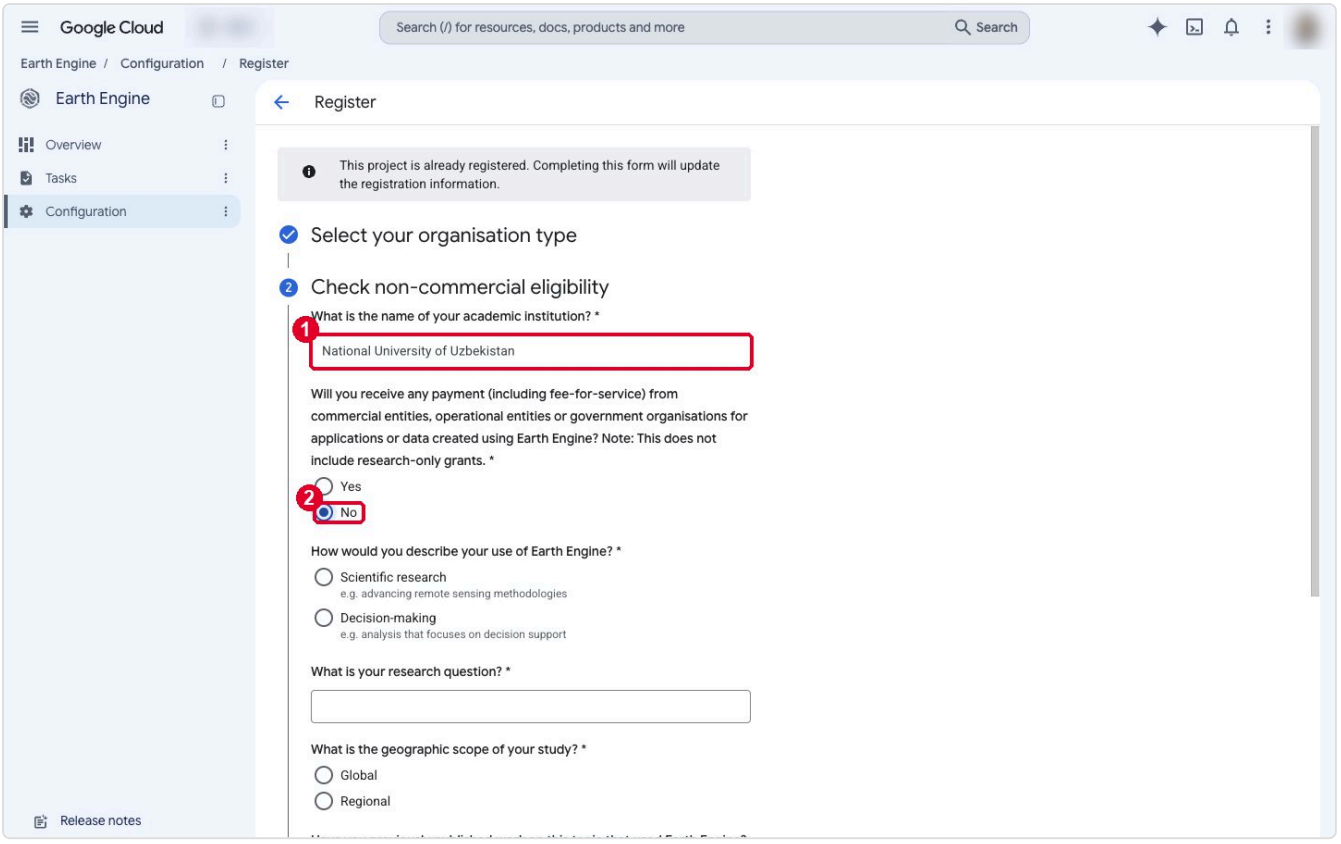
On the Earth Engine page, click **Register**. A form with five steps opens.

Step 1: Select your organisation type. Open the list (**1**), choose **Public or private academic institution (including department, staff, students)**, and click **Next** (**2**). Teachers and trainers can also choose **Earth Engine trainer or trainee**.

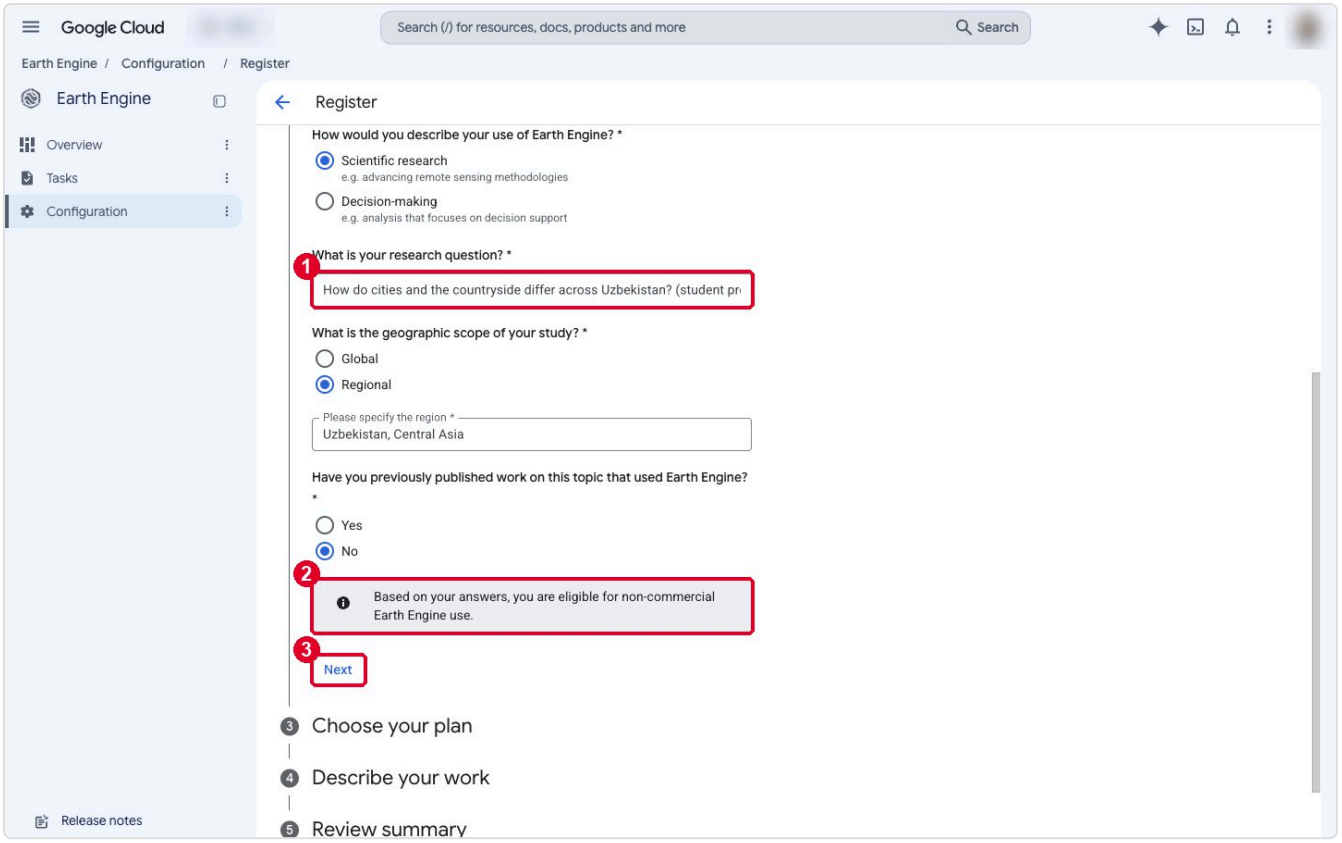


Step 2: Check non-commercial eligibility.

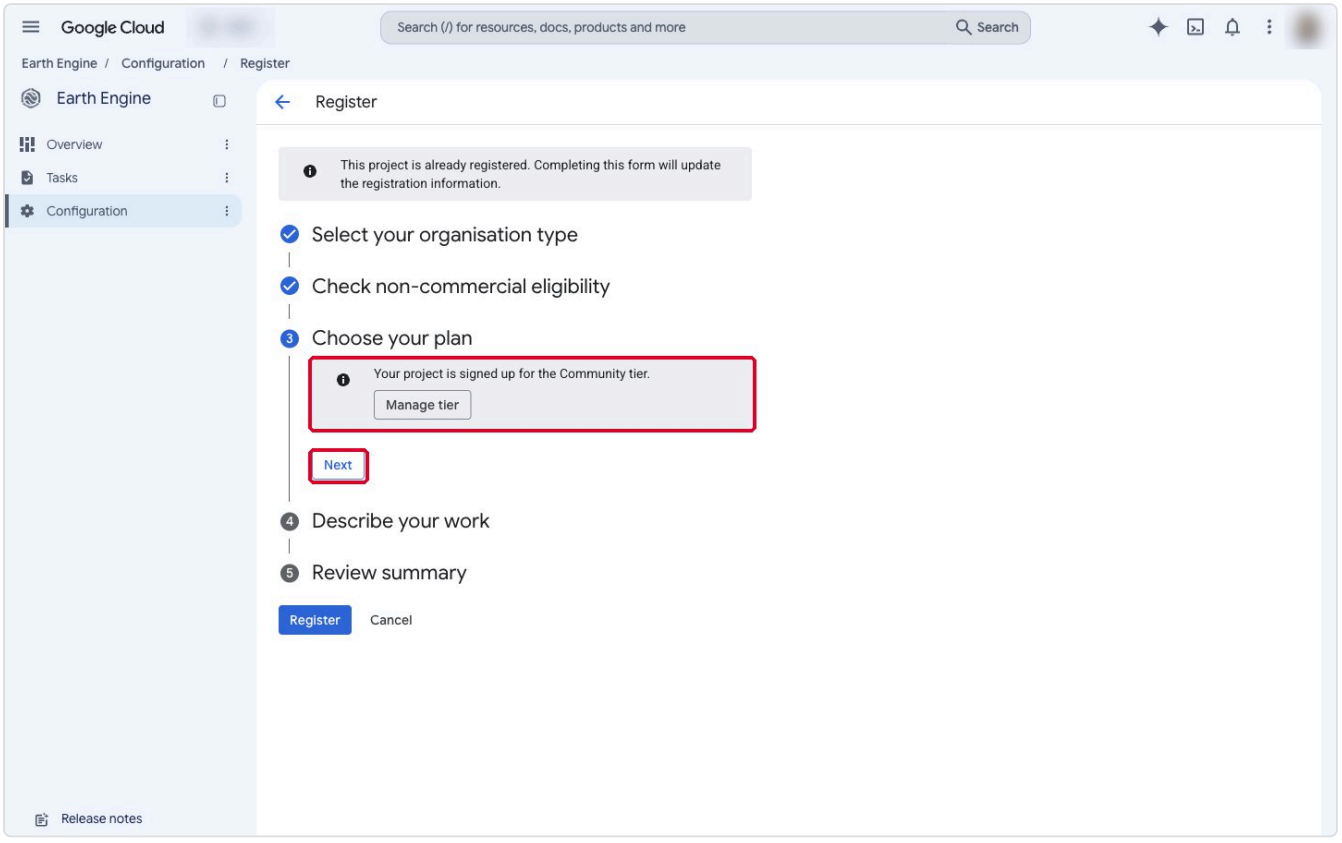
- Type the name of your university or school (**1**).
- *Will you receive any payment … from commercial entities?* Choose **No** (**2**).



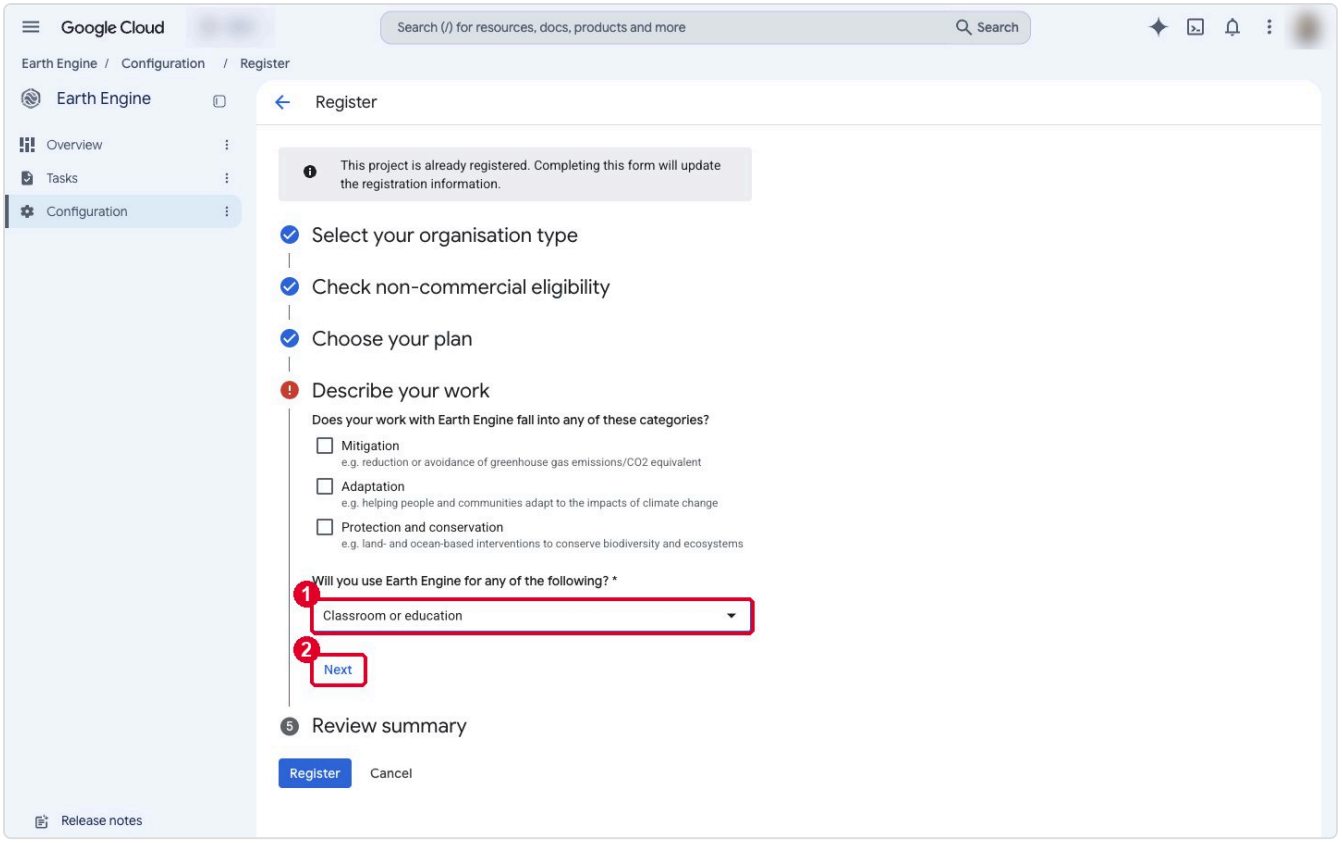
- *How would you describe your use of Earth Engine?* **Scientific research**.
- *What is your research question?* Write one sentence about your project (**1**), for example: *How do cities and the countryside differ across Uzbekistan?*
- *Geographic scope:* **Regional**, and type **Uzbekistan, Central Asia**.
- *Have you previously published work …?* **No**.
- Click **Check eligibility**. When the message says **you are eligible for non-commercial Earth Engine use** (**2**), click **Next** (**3**).



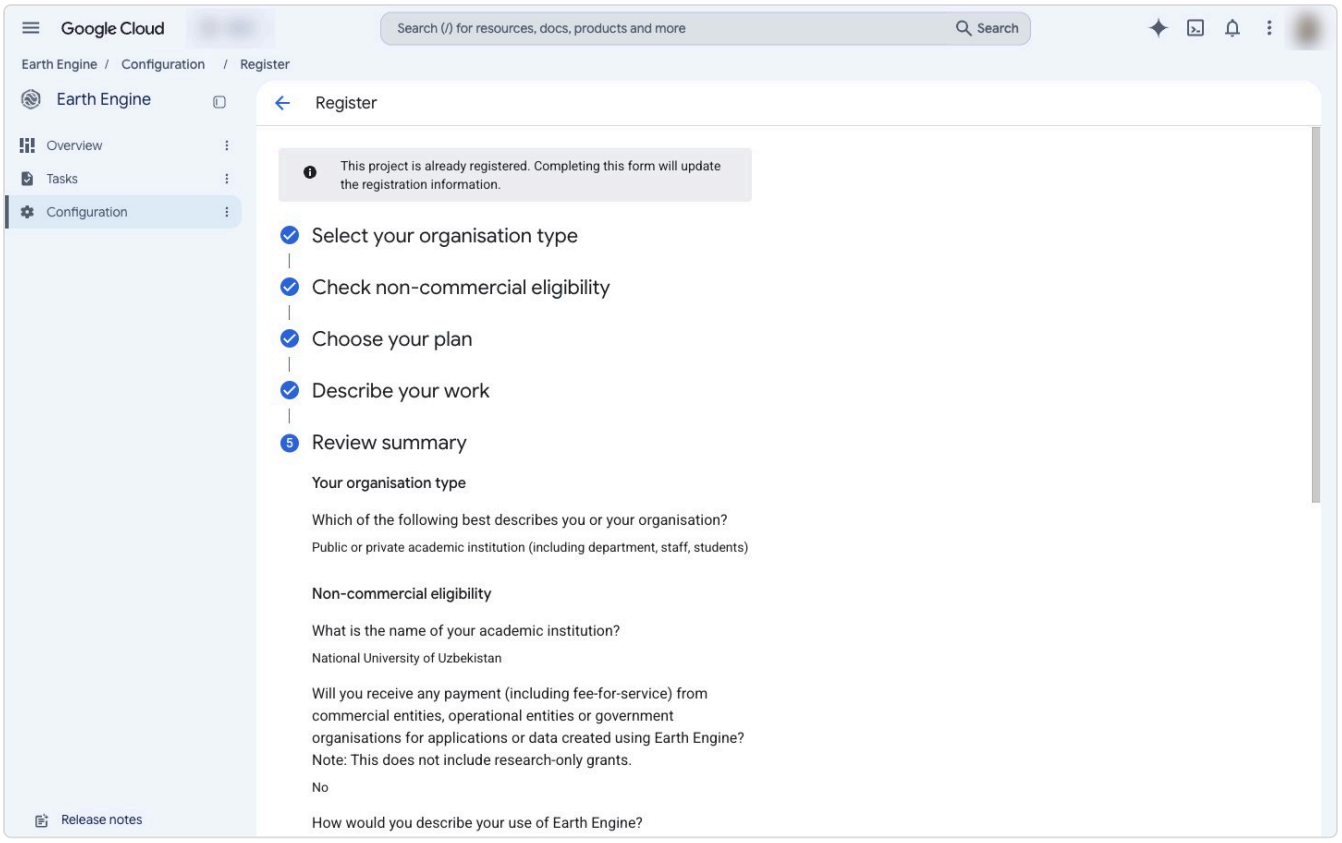
Step 3: Choose your plan. Choose the free **Community tier** (it may already be chosen for you), then click **Next**.



Step 4: Describe your work. You can leave the three checkboxes empty. In *Will you use Earth Engine for any of the following?* open the list, tick **Classroom or education**, click **OK** (**1**), then **Next** (**2**).



Step 5: Review summary. Check your answers and click the blue **Register** button at the bottom.



> Our screenshots were taken with a project that was already registered, so they show a grey note at the top, *This project is already registered*. You won't see it on a new project.

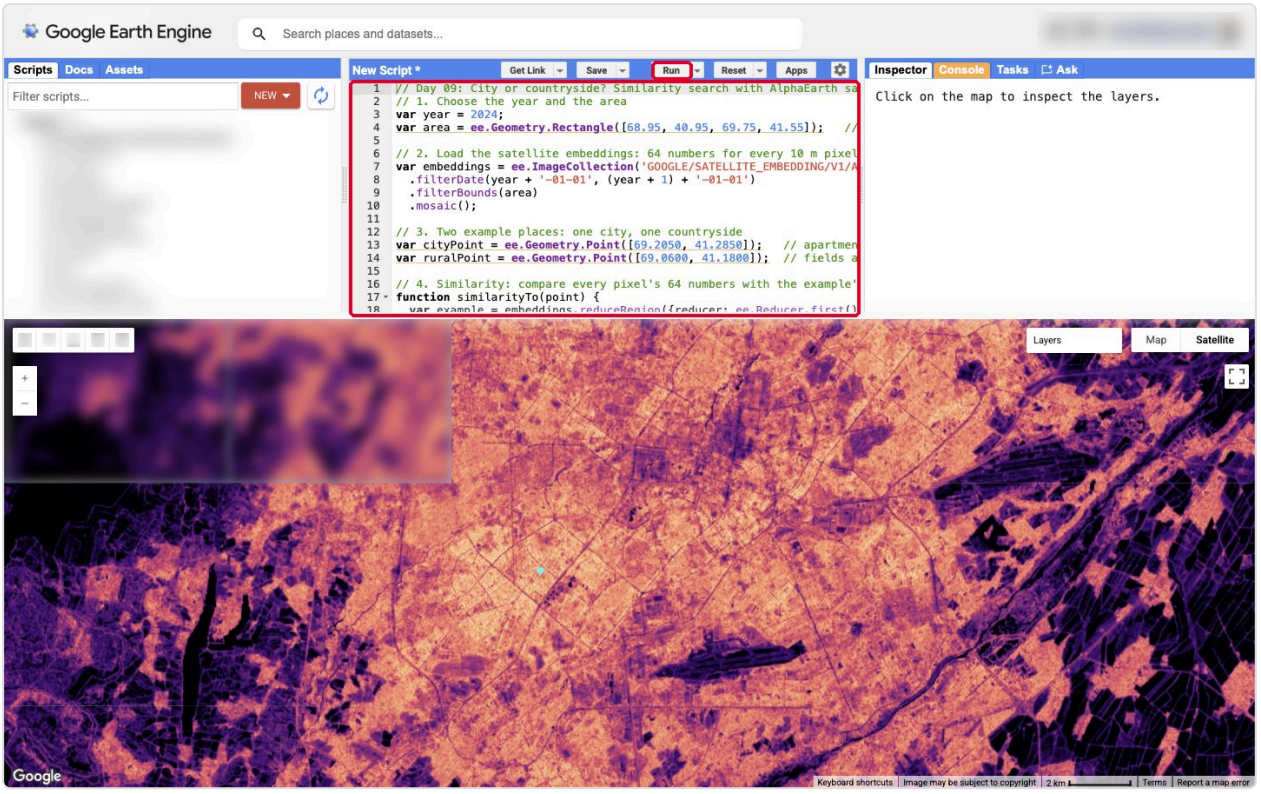
1d. Open the Code Editor

Go to **code.earthengine.google.com**. If it asks for a project, choose the one you just registered. The **Code Editor** opens: scripts on the left, the code in the middle, the console on the right and the map at the bottom.

> **Free, but not unlimited.** The Community Tier gives every noncommercial project a monthly amount of computing time. This tutorial uses only a small part of it. If you see a yellow message about quota, the map still works, just more slowly.

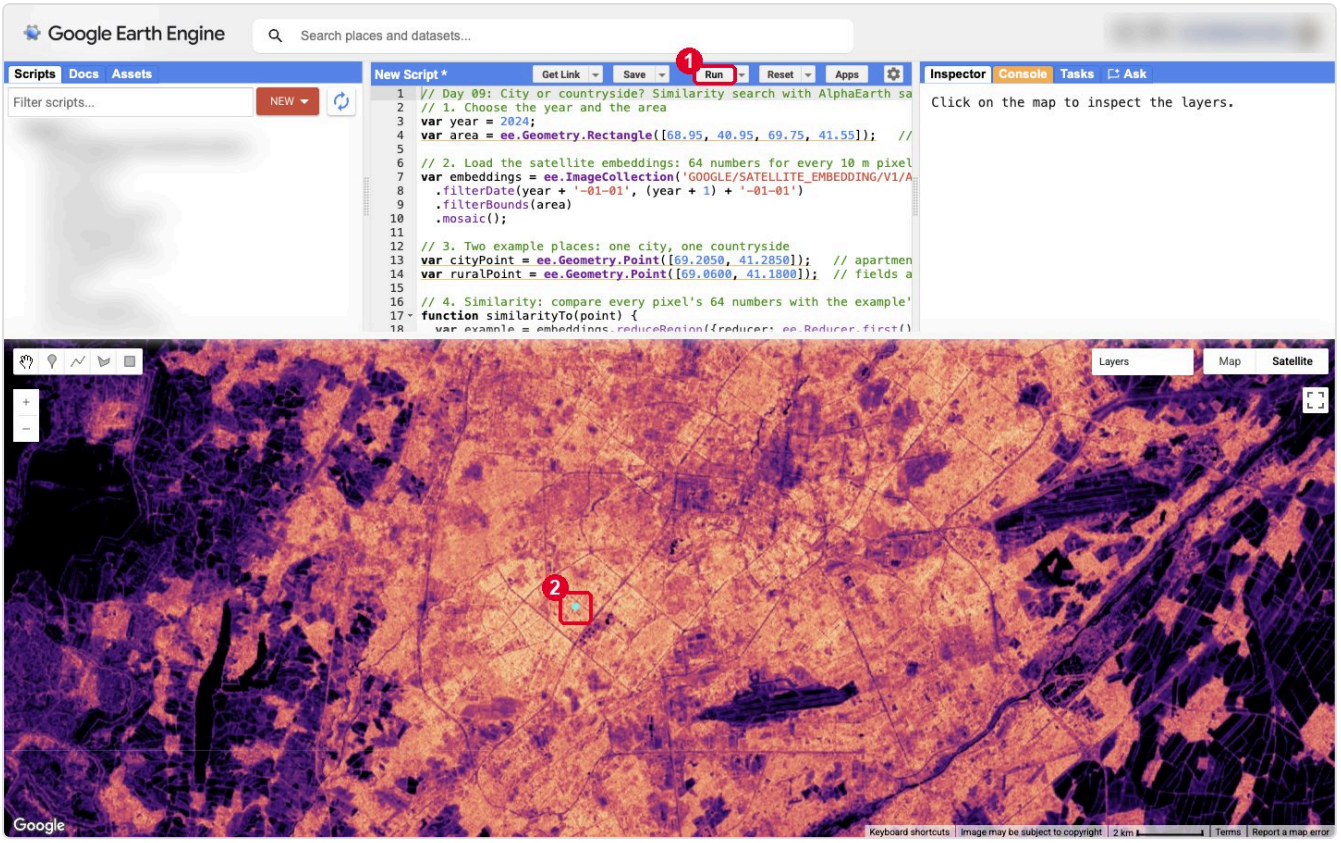
Part 2. Paste the script and run it

1. Open our script: **day09_similarity.js**. Click **Copy raw file** (the two-squares button) on GitHub.
2. Click into the empty middle panel of the Code Editor and paste (**Ctrl+V**, or **⌘V** on a Mac).
3. Click **Run**.



After a few seconds the map flies to Tashkent. The **Console** on the right says **Embedding bands: 64**: the 64 numbers of each pixel.

The colourful layer is **Similar to the city** (**2**). The small blue dot is our city example, apartment blocks in Chilonzor. Bright yellow and orange mean "very similar to this place"; dark purple means "very different".



Look how clearly the city stands out. The dark shapes inside it are the things that aren't "city": parks, the Ankhor canal, the lakes and the airport's runways.

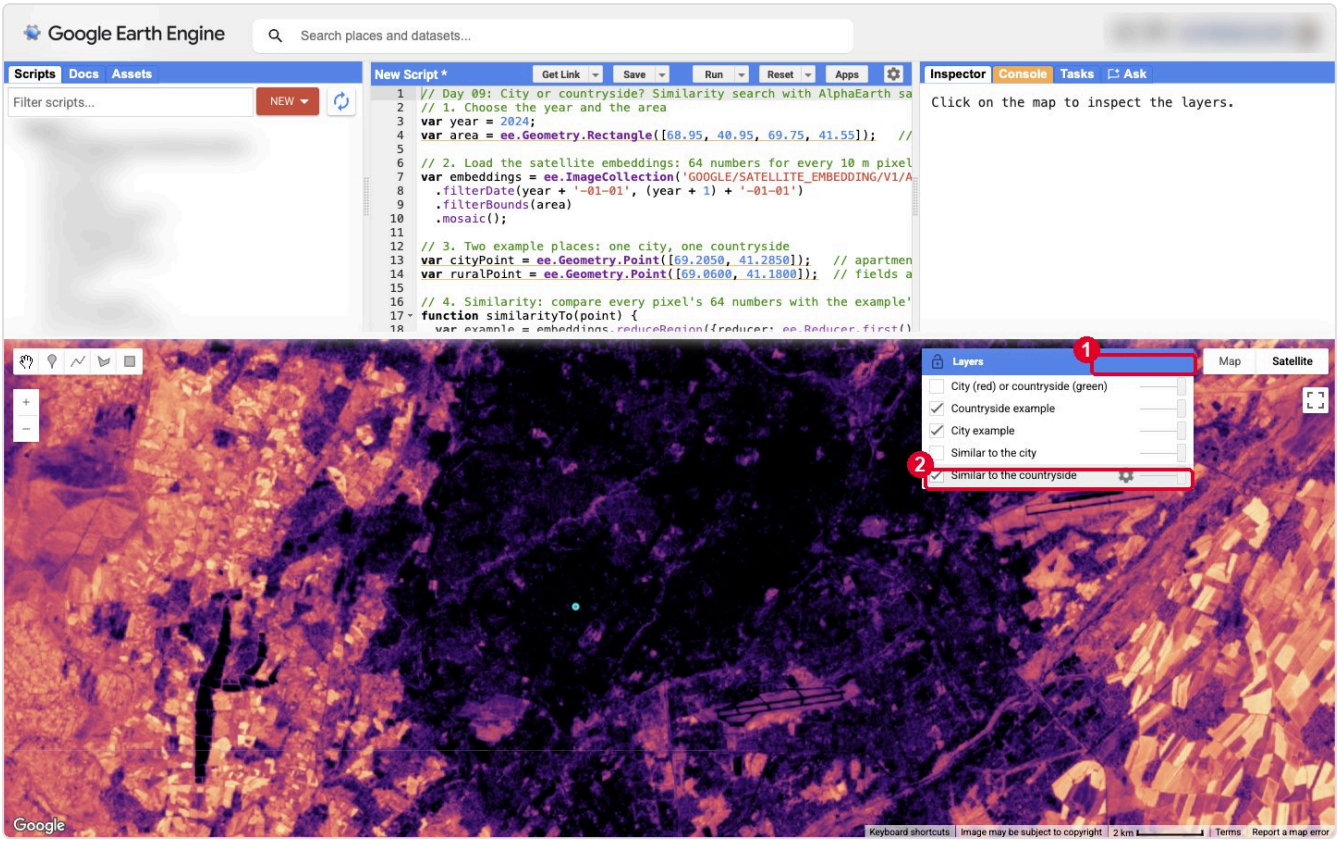
Part 3. What the script does, line by line

Part of the script	What it does
`var area = …Rectangle(…)`	The area we look at: Tashkent and the farmland around it.
`ee.ImageCollection('GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL')`	The AlphaEarth Satellite Embedding dataset: one image with 64 bands for each year.
`.filterDate(year…)` and `.mosaic()`	Keeps only the year we want (2024) and joins the tiles into one image.
`cityPoint`, `ruralPoint`	The two example places, as longitude and latitude.
`similarityTo(point)`	Reads the 64 numbers of the example, multiplies them with the 64 numbers of every pixel and adds them up: the dot product.
`Map.addLayer(…)`	Draws the result on the map with a colour palette.
`cityLike.gt(ruralLike)`	For every pixel: is it more like the city (red) or more like the countryside (green)?
`Map.onClick(…)`	Lets you click anywhere and search for places like that one.

Part 4. Switch to the countryside

1. Click **Layers** at the top right of the map (**1**).

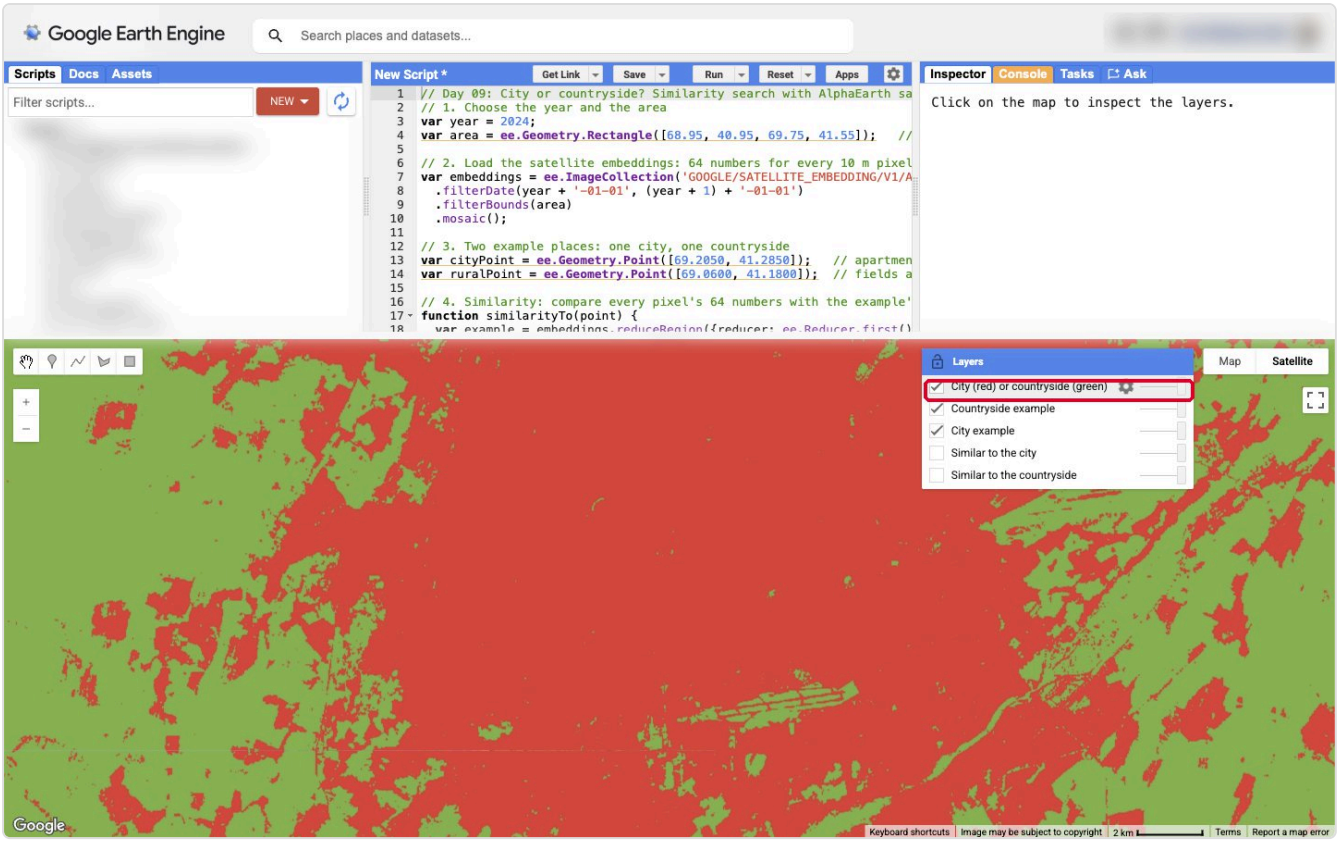
2. Untick **Similar to the city** and tick **Similar to the countryside** (**2**).



Now it's the other way round: the city goes dark and the fields around it light up. The countryside example is a cotton and wheat field south-west of the city, outside this view.

Part 5. City or countryside, for every pixel

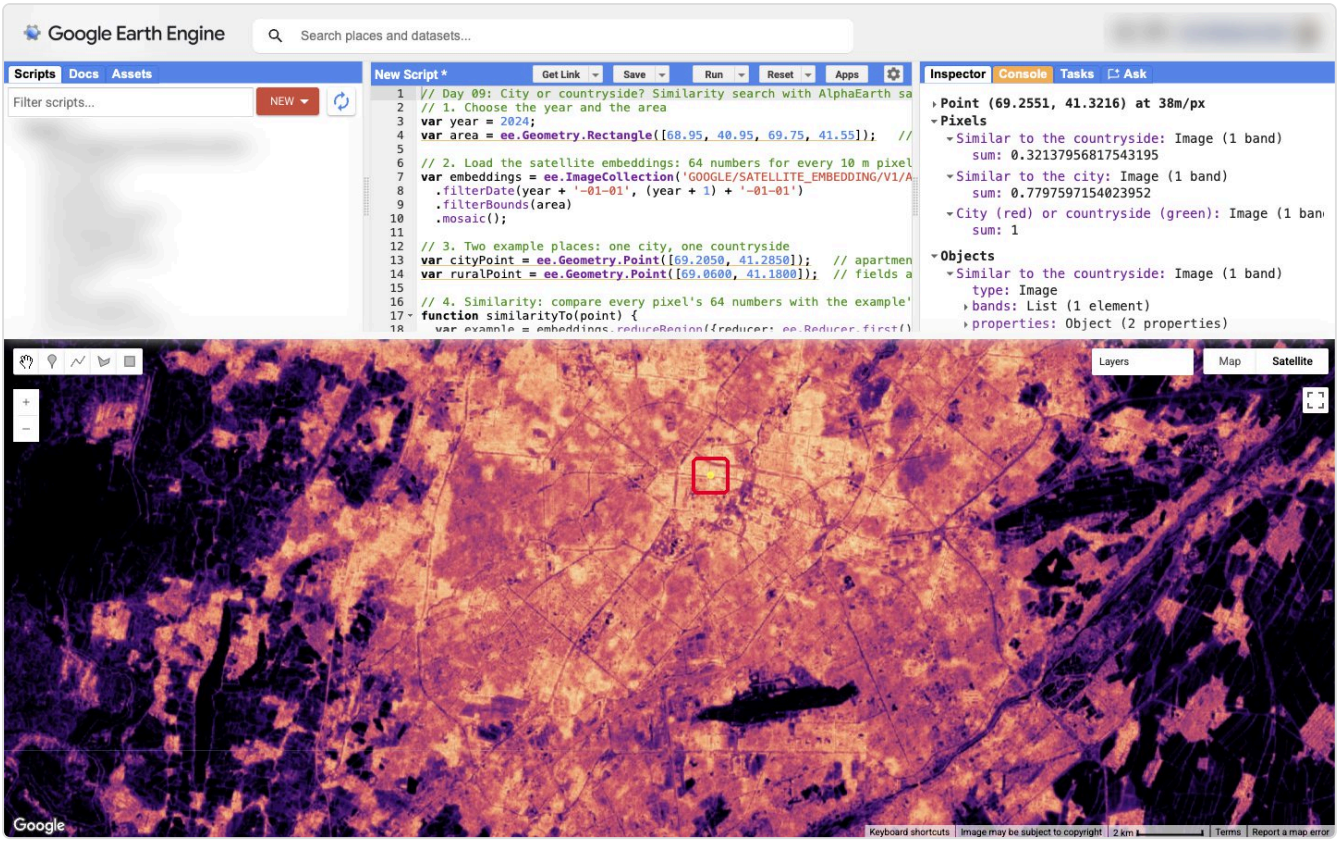
In **Layers**, untick the similarity layer and tick **City (red) or countryside (green)**. Each pixel is red if it's more similar to the city example, and green if it's more similar to the field.



With only two examples this is a rough map: the countryside around Tashkent is full of villages and gardens, and some of them look more like the city than like a field. That's the interesting part of "urban–rural": the line between them isn't sharp.

Part 6. Your turn: click any place

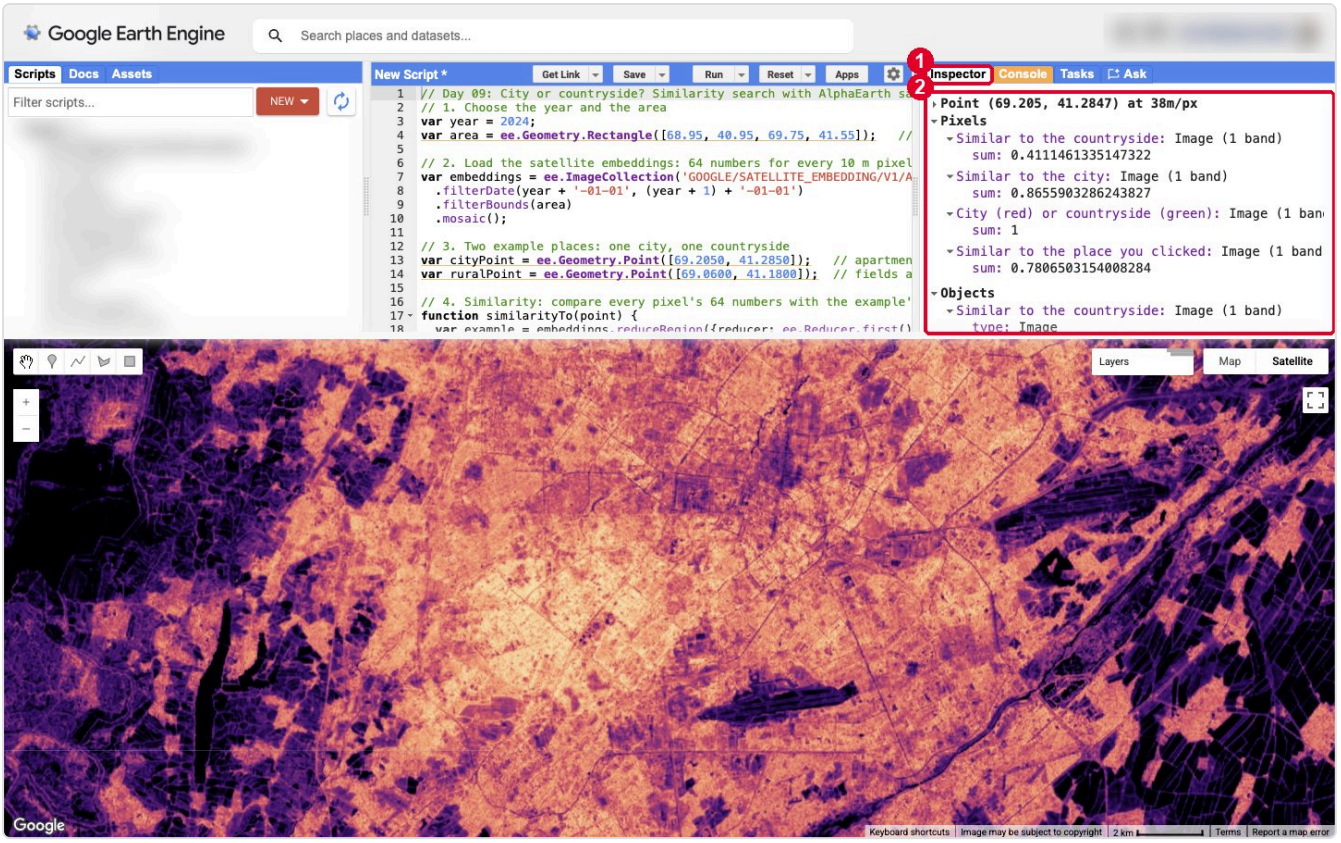
1. Close the **Layers** panel. The cursor on the map is now a cross.
2. Click any place you're curious about: your school, a mahalla with houses and gardens, a bazaar, a park.
3. Wait a few seconds. A new layer appears: **Similar to the place you clicked**. A yellow dot marks your place, and the **Console** tab prints its coordinates.



Here we clicked a spot near Abay Street in the city centre (Urda). Places like it light up across the whole centre, while the fields, the canals and the big parks stay dark. Try a mahalla with houses and gardens next: you'll see a different pattern.

Part 7. Read the numbers with the Inspector

1. Click the **Inspector** tab at the top right (**1**).
2. Click a place on the map. The Inspector lists the value of every layer at that point (**2**). For example, at our city example **Similar to the city** is 0.87, but **Similar to the countryside** is only 0.41.



Part 8. Save and share

- **Save:** click **Save** above the code and give the script a name. It's stored in your own scripts list on the left.
- **Share:** click **Get Link**. Earth Engine makes a link to a copy of your script; anyone who opens it can run it. Be careful: the link is public.
- **Picture for your post:** make the map full screen with the square button on the right of the map, hide the panels and take a screenshot.

Ideas for your own map

- **Another city.** Change the area and the two example points to Samarkand, Namangan or Nukus.
- **Another year.** Change `var year = 2024;` to 2017 and compare. Which fields around Tashkent became city?
- **More than two examples.** Add a third example, for example a mahalla, and colour each pixel by the example it's most similar to.
- **Find a type of place.** Click one greenhouse, one solar farm or one orchard, and find all the others in the Fergana Valley.

When you share

Please credit:

> The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind (CC BY 4.0). Map made with Google Earth Engine.

Troubleshooting

Problem	What to do
Earth Engine asks for a project.	Register one (Part 1), then choose it from the project menu at the top right.
Run does nothing, or there's a red error in the Console.	Make sure you pasted the whole script, from the first line to the last `});`.
The map stays grey or blurry for a long time.	Wait: the first view can take up to a minute. Zoom in a little to compute fewer pixels.
A yellow message says the quota is exceeded.	Your project has used its monthly computing time. The map still works, but slowly. Use a smaller area, or wait until next month.
Everything is bright, or everything is dark.	Change `min: 0.4, max: 1` in `Map.addLayer` to show a narrower range, for example `min: 0.7`.

Credits

- **Satellite embeddings:** The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind. Licensed CC BY 4.0. Earth Engine ID: `GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL`, 2017–2024, 10 m.
- **Method:** based on Google's similarity search tutorial for the Satellite Embedding dataset.
- **Software:** Google Earth Engine Code Editor.